

Notes: Chen & Kung (QJE, 2019)
**Busting the “Princelings”: The Campaign against Corruption in China’s
Primary Land Market**

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	3
2.1	Institutional background: why the primary land market is the right laboratory . . .	3
2.2	Identifying the princelings and their firms	4
2.3	Characteristics of princeling firms	4
2.4	Land transaction data and the matched sample	4
2.5	Main variables	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Conceptual framework (no formal structural model)	5
3.2	Main land-price regression	5
3.3	Heterogeneity by the patron’s political power and retirement	6
3.4	Corresponding quantity regression	6
3.5	Political-turnover regression	7
3.6	Refined measures of favor: discount size and land area	7
3.7	Anticorruption interaction regressions	7
3.8	Event-study design around Zhou Yongkang’s arrest	8
4	Identification and empirical design (what makes the estimates credible)	8
4.1	Why the land-price estimates are the paper’s cleanest evidence	8
4.2	Supportive patterns that reinforce the corruption interpretation	9
4.3	How the paper addresses data-quality concerns	9
4.4	Why the promotion results are more suggestive than the pricing results	9
4.5	Why the anticorruption evidence strengthens causal interpretation	9

5	Tables and figures	9
5.1	Figure I: Size of Land Revenue, 2000–2016	9
5.2	Figure II: The power pyramid of the CCP	10
5.3	Tables I and II: Who the princelings are and what their firms look like	10
5.4	Figure III and Table III: Geographic spread and the matched-sample design	11
5.5	Table IV: Raw comparison of land parcels purchased by connected and unconnected firms	12
5.6	Table V: Princeling purchase and land price	12
5.7	Figure IV: Scatter of princeling and nearby nonprinceling prices	12
5.8	Table VI: Quantity of land purchased by princeling firms	13
5.9	Table VII: Summary statistics on political turnover	13
5.10	Table VIII: Princeling purchase and provincial leaders' promotion	13
5.11	Table IX: Princeling purchase and municipal leaders' promotion	14
5.12	Table X and Figure V: Xi's campaign and the decline in price discounts	15
5.13	Table XI and Figure VI: Xi's campaign and the quantity of land transferred	15
5.14	Table XII and Figure VII: Xi's campaign and the collapse of the promotion channel	16
6	Short summary	16
7	Conclusion	17
7.1	Core conclusion (what the paper establishes)	17
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** How does cronyism work in an autocratic system where local officials control valuable state assets but promotion is decided from above? The paper studies whether Chinese local officials sold land cheaply to firms connected to Politburo families (“princelings”) and whether they were rewarded with promotion.
- **Core mechanism.** In this setting, the local official is effectively the party who provides the bribe: he or she controls land-use rights, can price-discriminate across buyers, and may sell land at a discount to politically connected firms in exchange for future political advancement.
- **Institutional ingredients.** Two features matter: (i) decentralization of control rights over land and other state assets to local governments, and (ii) the “one-level-up” promotion rule, under which higher-ranked party elites control appointments and promotions of lower-ranked officials.
- **Setting.** China’s primary land market, where local governments are the sole statutory sellers of land-use rights. The paper studies more than one million land transactions from 2004–2016 and links them to firms connected to the families of Politburo and Politburo Standing Committee members during 1997–2016.
- **Main finding on rents.** Firms linked to Politburo families obtain extremely large land-price discounts. Depending on the comparison sample, the estimated discount is about 55.4% in the full sample and about 57.0% to 59.5% in close spatially matched samples.
- **Power gradient.** Connections to Politburo Standing Committee members are even more valuable: those firms receive an additional 17.6% to 20.7% discount beyond the discount associated with an ordinary Politburo connection.
- **Main finding on reciprocity.** Provincial party secretaries who provide these favors are more likely to be promoted. The paper’s headline estimate is that princeling-related land deals raise their promotion probability by about 23.4%. The same is not true for provincial governors.
- **Main finding on anticorruption.** Xi Jinping’s anticorruption campaign materially reduces these discounts. In the matched sample, central inspections reduce corruption by about 42.6%, and replacing a provincial party secretary with a Xi-appointed official reduces it by about 31.5%.
- **Why the paper matters.** The paper gives a direct, transaction-level measure of political rents rather than inferring them from stock prices or firm valuation. It also shows how corruption is embedded in promotion incentives and how monitoring plus personnel control can curb it.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: why the primary land market is the right laboratory

- **Primary land market.** Local governments are the only legal sellers of land-use rights. This makes the market close to the monopoly setting emphasized in the corruption literature: one public seller, many private buyers, and ample room for price discrimination.
- **Fiscal importance.** Land sales became a central source of local public finance. The paper emphasizes that land revenue was less than 2% of county-level total revenue in 2000 but rose to about 30% by 2011, amounting to roughly 6% of GDP.

- **Political hierarchy.** Promotion is controlled by the level above. For province-level leaders, Politburo members matter directly, so local officials have strong incentives to cultivate the families of those elites.
- **Why corruption should show up here.** Land is valuable, administratively allocated, and observable at the transaction level. That lets the authors compare the price paid by connected and unconnected firms for similar parcels.

2.2 Identifying the princelings and their firms

- **Political elites.** The authors identify 64 Politburo members across the 15th to 18th Party Congresses, spanning 1997–2016. Of these, 36 are ordinary Politburo members and 28 are Politburo Standing Committee members.
- **Family members.** They identify 134 princelings connected to 48 of those elites using multiple public sources (major Western media, Chinese news groups, and the International Consortium of Investigative Journalists).
- **Relationship mix.** About 58.2% are sons, daughters, or children-in-law; about 23.1% are wives or other in-laws; the rest are siblings, nephews, or nieces.
- **Occupation mix.** About 45.5% are private firm owners or investors, 29.1% are SOE executives, and the rest are government officials, military officials, or other occupations.
- **Connected firms.** The paper identifies 3,530 firms (including subsidiaries) linked to princelings. Those firms make up only 0.81% of all land-buying firms but account for 1.71% of total land revenue, or about 4.3 trillion yuan in transactions.

2.3 Characteristics of princeling firms

- **Sectoral concentration.** Princeling firms are heavily concentrated in real estate (36.67%) and finance (18.70%), exactly the sectors where access to land and state assets is especially valuable.
- **Ownership and size.** Compared with nonprinceling firms, princeling firms are less likely to be state owned (23.64% versus 34.66%) but much more likely to be large (72.61% versus 41.37%).
- **Location of headquarters.** Roughly 65.29% of princeling firms are headquartered in a national or provincial capital, versus only 24.57% for nonprinceling firms.
- **Interpretation.** These are not fringe firms. They are large, politically connected, and concentrated in exactly the activities where local officials can generate rents.

2.4 Land transaction data and the matched sample

- **Universe of transactions.** The land dataset covers 1,628,635 primary-market transactions between 2004 and 2016. Of these, 1,151,358 involve firms rather than private individuals.
- **Transaction details.** Each observation includes price, parcel size, exact location, date, seller, buyer, sales method, land-use type, land quality (rated on a 20-point scale), and the buyer firm's industry.
- **Princeling purchases.** Connected firms purchase 19,812 land parcels during the sample period.

- **Matched comparison groups.** To compare like with like, the paper matches nonprinceling transactions within a 500-meter or 1,500-meter radius of a princeling parcel in the same year. The 1,500-meter sample contains 359,539 nonprinceling parcels; the 500-meter sample contains 207,564.
- **Why the spatial match matters.** It is designed to reduce selection bias from connected firms buying especially desirable parcels. Within a narrow radius, land quality and local market conditions should be much more comparable.

2.5 Main variables

- **Outcome 1: land price.** The main land-market outcome is log price per square meter.
- **Outcome 2: promotion.** Political turnover is coded ordinally: termination = 0, retirement = 1, lateral transfer or staying in office = 2, promotion = 3.
- **Treatment.** PP is a dummy equal to one when the purchasing firm is connected to a princeling.
- **Additional connection measures.** The paper distinguishes connections to Politburo Standing Committee members, retirement status of the patron, and refined province-year or municipality-year measures based on the size of discounts and land area sold.
- **Campaign variables.** The main anticorruption indicators are: a post-2012 dummy, a central-inspection dummy, and a dummy for provinces whose party secretary was newly appointed by Xi.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework (no formal structural model)

- The paper’s mechanism is not built around a formal equilibrium model. Instead, it is an institutional argument: local governments control land, higher-ranked elites control promotion, and princeling firms sit at the intersection of those two hierarchies.
- The implied exchange is: cheap land now in return for political help later. The empirical strategy is therefore to look directly for discounted prices, larger quantities sold, and higher subsequent promotion rates of the officials who provide those favors.

3.2 Main land-price regression

The core transaction-level specification is

$$Price_{ickst} = \beta_0 + \beta_1 PP_{ikjt} + \gamma X_i + T_{cst} + \nu_{ickst}. \quad (1)$$

- **Dependent variable.** $Price_{ickst}$ is the log price per square meter for parcel i , sold by municipality c to firm k , designated for use s , in month-year t .
- **Regressor of interest.** $PP_{ikjt} = 1$ if buyer k is connected to princeling j .

- **Controls.** X_i includes land quality, log land area, sales-method dummies, firm size, and ownership.
- **Fixed effects.** T_{cst} stands for high-dimensional city-year-usage fixed effects; the regressions also include month fixed effects and three-digit industry fixed effects.
- **Inference.** Standard errors are two-way clustered by province and firm.
- **How to read β_1 .** A negative β_1 means connected firms pay less for otherwise comparable land.

Because the dependent variable is in logs, the paper converts coefficients to percentage discounts using

$$\text{Discount} = \exp(\beta_1) - 1. \quad (2)$$

- **Example.** A coefficient of -0.844 implies $\exp(-0.844) - 1 \approx -0.57$, that is, about a 57% discount.

3.3 Heterogeneity by the patron's political power and retirement

The main price equation is extended to allow the value of connections to vary with the patron's rank and status:

$$\begin{aligned} Price_{ickst} = & \beta_0 + \beta_1 PP_{ikjt} + \beta_2 (PP_{ikjt} \times PSCM_{jp}) \\ & + \beta_3 (PP_{ikjt} \times Retired_{jpt}) + \gamma X_i + T_{cst} + \nu_{ickst}. \end{aligned} \quad (3)$$

- **PSCM.** Indicates that the connected patron is a Politburo Standing Committee member rather than an ordinary Politburo member.
- **Retired.** Indicates the transaction occurs after that patron's retirement.
- **Interpretation.** $\beta_2 < 0$ means connections to more powerful elites buy even steeper discounts. $\beta_3 \approx 0$ means the rents survive retirement.

3.4 Corresponding quantity regression

To test whether connected firms also receive more land, the paper estimates a firm-year quantity regression of the form

$$\begin{aligned} \log(\text{AreaPurchased}_{ft}) = & \alpha_0 + \alpha_1 \text{PrincelingFirm}_f + \alpha_2 (\text{PrincelingFirm}_f \times \text{PSCM}_f) \\ & + \alpha_3 (\text{PrincelingFirm}_f \times \text{Retired}_{ft}) + Z_{ft}\Pi + \eta_{industry} + \tau_t + u_{ft}. \end{aligned} \quad (4)$$

- **Dependent variable.** Log annual area purchased by firm f .
- **Interpretation.** This is the quantity analogue of the price regression: do connected firms buy more land, not just cheaper land?
- **Expected sign.** $\alpha_1 > 0$ if connections expand access as well as reduce price.

3.5 Political-turnover regression

The paper then asks whether local officials who provide favors are rewarded with promotion. The core specification is

$$\begin{aligned} Turnover_{it} = & \phi_0 + \phi_1 PP_{it} + \phi_2 FractionalTies_{jt} + \phi_3 GDPGrowth_{it} \\ & + \kappa X_{it} + \omega W_j + \lambda_i + T_t + \varphi_j + \tau_{ijt}. \end{aligned} \quad (5)$$

- **Dependent variable.** Ordered political turnover of a provincial or municipal official in year t .
- **Estimation.** The paper mainly uses ordered probit, with a binary promotion-versus-no-promotion specification as a robustness check.
- **Controls.** Economic performance (GDP growth or tax-revenue growth), factional ties, log GDP per capita, log population, age, age squared, and years of schooling.
- **Interpretation.** $\phi_1 > 0$ means officials who helped princeling firms are more likely to move upward.
- **Important caveat.** The paper is explicit that political connections are endogenous here and there is no clean technical fix. The promotion regressions are best read as strong evidence of reciprocity, but less cleanly causal than the spatially matched price estimates.

3.6 Refined measures of favor: discount size and land area

For the promotion regressions, the paper supplements the simple purchase dummy with two richer province-year or municipality-year measures:

$$PD_{pt} = \sum_{i \in (p,t)} \max\left\{0, Price_i^{match} - Price_i^{princeling}\right\}, \quad (6)$$

and

$$ALP_{pt} = \log(1 + \text{total area sold to princeling firms in } (p, t)). \quad (7)$$

- **Interpretation.** PD measures how large the discount is in the aggregate; ALP measures how much land is transferred to connected firms.
- **Why useful.** These variables ask not just whether an official dealt with a princeling firm, but how much favor was extended.

3.7 Anticorruption interaction regressions

To measure the effect of Xi's campaign on transaction prices, the authors augment the price equation with campaign interactions:

$$\begin{aligned} Price_{ickst} = & \beta_0 + \beta_1 PP_{ikjt} + \beta_2 (PP_{ikjt} \times Campaign_{ct}) \\ & + \beta_3 Campaign_{ct} + \gamma X_i + T_{cst} + \nu_{ickst}. \end{aligned} \quad (8)$$

- **Campaign variable.** Alternately defined as: transaction after 2012, central inspection, Xi-appointed provincial party secretary, or pre-2012 anticorruption inspection.
- **Interpretation.** $\beta_2 > 0$ means the campaign weakens the price discount previously enjoyed by princeling firms.
- **Economic logic.** More monitoring and tighter personnel control should reduce a local official's willingness to trade land rents for promotion.

The quantity analogue is

$$\log(Area_{fpt}) = \delta_0 + \delta_1 PrincelingFirm_f + \delta_2 (PrincelingFirm_f \times Campaign_{pt}) + \delta_3 Campaign_{pt} + Z_{fpt}\Lambda + \mu_p + \tau_t + \varepsilon_{fpt}. \quad (9)$$

- **Interpretation.** $\delta_2 < 0$ means the campaign reduces the extra quantity of land going to connected firms.

3.8 Event-study design around Zhou Yongkang's arrest

The paper's final design is graphical rather than regression-heavy: it tracks daily prices and daily quantities of land transactions by princeling and nonprinceling firms around December 1, 2013, immediately after the news of Zhou Yongkang's investigation.

$$Gap_t = Y_t^{princeling} - Y_t^{nonprinceling}, \quad t \in \text{days around Dec. 1, 2013}. \quad (10)$$

- **Interpretation.** If local officials suddenly fear scrutiny, the price gap should shrink and the quantity purchased by princeling firms should fall right after the arrest news.
- **Role in the paper.** This is a sharp narrative shock that complements the broader campaign evidence.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the land-price estimates are the paper's cleanest evidence

- The strongest part of the paper is the transaction-level pricing exercise. It compares parcels within the same city, land-use category, month-year, and buyer industry, while also controlling for land quality, parcel size, sales method, firm size, and ownership.
- The spatially matched samples are especially persuasive. Comparing a connected parcel with nearby unconnected parcels within 500 or 1,500 meters in the same year sharply narrows differences in underlying land quality and local demand conditions.
- The fact that the estimated discount gets even larger in the close-match samples is informative: it suggests that selection likely biased the full-sample estimate downward because princeling firms buy better parcels.

4.2 Supportive patterns that reinforce the corruption interpretation

- More powerful patrons generate larger discounts. That is exactly what one would expect if political clout rather than ordinary business relationships drives the result.
- The discount does not disappear after the patron retires. That is consistent with the continuing influence of retired Chinese elites.
- Connected firms buy not just cheaper land, but slightly more land, which is exactly the quantity-side implication of favoritism.
- The discount is much larger for commercial/residential land than for industrial land, consistent with commercial land being the place where the highest rents are available.

4.3 How the paper addresses data-quality concerns

- One concern is that media-identified connections may be incomplete or noisy. The paper addresses this by re-running the analysis on two cleaner subsamples: firms revealed by the Zhou Yongkang case and publicly listed connected firms.
- The main price-discount result survives in both samples. That is important because it makes it less likely that the baseline findings are an artifact of selective reporting.

4.4 Why the promotion results are more suggestive than the pricing results

- The promotion regressions control for GDP growth, tax-revenue growth, factional ties, and leader characteristics, and they distinguish party secretaries from governors or mayors.
- Still, promotion is inherently political and connections are endogenous. The paper itself notes that there is no quick econometric fix here.
- For that reason, the most credible reading is: the promotion results strongly support the reciprocity story, but they are not as cleanly identified as the parcel-price estimates.

4.5 Why the anticorruption evidence strengthens causal interpretation

- Central inspections and Xi-appointed provincial leaders are plausibly sharper changes in monitoring and personnel control than a simple post-2012 dummy.
- The fact that pre-2012 inspections do not reduce discounts, while Xi-era inspections do, supports the idea that the post-2012 campaign was qualitatively different rather than just more rhetoric.
- The Zhou Yongkang event study gives an additional sudden shock. The immediate narrowing of the price gap after the arrest news is hard to reconcile with a slow-moving omitted variable.

5 Tables and figures

5.1 Figure I: Size of Land Revenue, 2000–2016

Read:

- The figure shows why land is such a valuable corruption margin. Land revenue rises dramatically after the mid-2000s and becomes a large share of both total government revenue and GDP.
- The key benchmark is that by 2011 land revenue is about 30% of total government revenue and roughly 6% of GDP.
- This figure is the fiscal backdrop for the entire paper: if local governments rely heavily on land, discretionary land sales become politically and financially central.

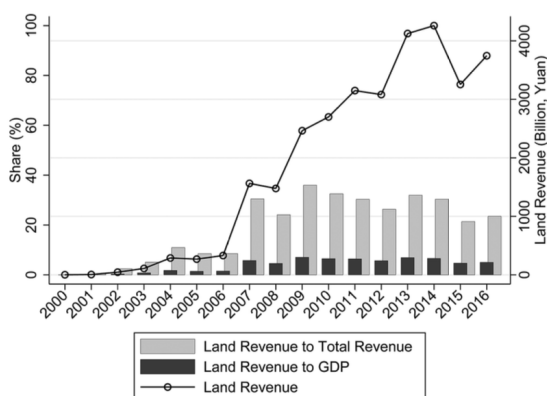


FIGURE I
Size of Land Revenue, 2000–2016

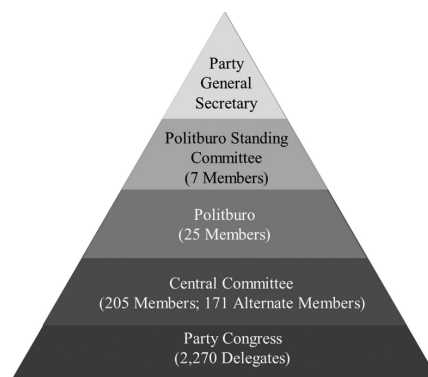


FIGURE II
The Power Pyramid of the 18th National Congress of the Communist Party of China (2012–2017)

5.2 Figure II: The power pyramid of the CCP

Read:

- This is the paper’s organizational chart: Party Congress at the bottom, then Central Committee, then Politburo, then Politburo Standing Committee, and finally the Party General Secretary.
- The point is to visualize why a connection to a Politburo family is so valuable. Those families are attached to the people who influence promotions at lower levels.
- The figure makes the “one-level-up” logic intuitive before any regression is run.

5.3 Tables I and II: Who the princelings are and what their firms look like

Read:

- Table I shows that the sample really is about close family ties, not vague acquaintances: most princelings are children or in-laws of top leaders.
- Table II shows that connected firms are large, often private, heavily concentrated in real estate and finance, and far more likely to be headquartered in political centers.
- Together these tables say that the paper is studying economically important firms and genuinely elite political connections.

TABLE I
CHARACTERISTICS OF PRINCELINGS

Total	134	
By relation to the Politburo and Politburo Standing Committee members		
Son/daughter and son-/daughter-in-law	78	58.21%
Wife and other in-laws	31	23.13%
Brother and sister	18	13.43%
Nephew and niece	7	5.22%
By occupation		
SOE high executives	39	29.10%
Private firm owners or investors	61	45.52%
Government officials	13	9.70%
Army officials	8	5.97%
Others	13	9.70%

TABLE II
SECTORAL CHARACTERISTICS OF PRINCELING FIRMS

Sector	Princeling firms		Nonprinceling firms	
	Number of firms	% of firms	Number of firms	% of firms
Real estate	1,294	36.67	116,829	26.69
Financial	660	18.70	4,985	1.14
Information technology and computer	391	11.08	11,067	2.53
Manufacturing	258	7.33	137,323	31.37
Electricity, gas, and water supply	222	6.31	19,061	4.35
Other	214	6.07	30,303	6.92
Leasing and business services	183	5.20	737	0.17
Environment and public facilities	105	2.99	371	0.08
Wholesale and retail trade	68	1.93	22,671	5.18
Transportation and storage	35	1.01	17,338	3.96
Accommodation and catering services	25	0.72	8,918	2.04
R&D	22	0.63	2,198	0.50
Agriculture	15	0.43	22,913	5.23
Service industry	13	0.39	37,696	8.61
Mining	8	0.24	1,758	0.40
News media and publisher	6	0.19	154	0.04
Education	1	0.05	1,069	0.24
Health and Social Services	1	0.05	2,385	0.54
Of which:				
State-owned firms	834	23.64	151,733	34.66
Large firms (annual revenue >0.3 billion yuan)	2,563	72.61	181,546	41.37
Headquartered in national/provincial capital	2,305	65.29	107,562	24.57
Total	3,530		437,776	

5.4 Figure III and Table III: Geographic spread and the matched-sample design

Read:

- Figure III shows that princeling transactions are geographically widespread but especially dense in eastern and coastal China and other major urban markets.
- The schematic under the map is crucial: the comparison is between a connected parcel and nearby unconnected parcels within a narrow radius.
- Table III confirms that the matched samples are not tiny side exercises; they contain hundreds of thousands of comparison transactions and similar average parcel sizes.

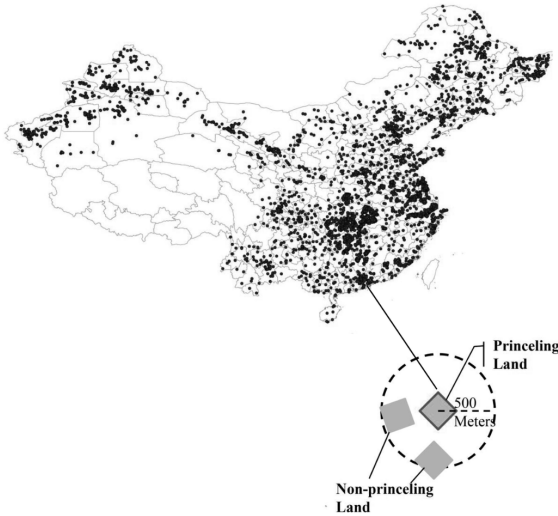


FIGURE III

Geographic Distribution of Land Transactions Conducted by Princeling Firms

TABLE III
SUMMARY STATISTICS OF LAND TRANSACTIONS

	All		≤ 1,500 meters		≤ 500 meters	
	Obs.	Mean	Obs.	Mean	Obs.	Mean
Log of land price	1,151,357	5.864	335,860	5.971	193,053	5.748
Princeling purchase	1,208,621	0.016	359,539	0.055	207,564	0.095
Land quality	1,208,621	12.969	359,539	13.876	207,564	13.448
Log of land area	1,184,522	4.813	349,929	4.691	201,515	4.690
Sales methods						
Bilateral agreement	226,910	0.188	82,733	0.230	44,905	0.216
English auction	99,854	0.083	29,096	0.081	14,655	0.071
Invited bidding	14,281	0.012	4,698	0.013	2,627	0.013
Listing auction	867,576	0.718	243,012	0.676	145,377	0.700

5.5 Table IV: Raw comparison of land parcels purchased by connected and unconnected firms

Read:

- This table makes the economic stakes visible even before regressions. Princeling firms overwhelmingly buy commercial/residential land, the high-rent category.
- In the matched 500-meter sample, princeling firms pay about 812 yuan per square meter for commercial/residential land versus about 1,457 yuan for comparable nonprinceling transactions.
- The industrial-land discount exists too, but it is much smaller. That helps localize the corruption margin to the most valuable parcels.

TABLE IV
LAND PARCELS PURCHASED BY PRINCELING AND NONPRINCING FIRMS, 2004–2016

	Number of transactions		Average price	
	Commercial and residential land	Industrial land	Commercial and residential land	Industrial land
All	586,974 (48.57%)	621,647 (51.43%)	1,383.99	280.28
Princelings' land	16,717 (84.38%)	3,095 (15.62%)	812.1	270.42
Nonprincelings' land	90,568 (48.24%)	97,184 (51.76%)	1,457.191	306.15
Difference			−645.09	−35.73

	Log of land price								
	All (1)	≤ 1,500 meters (2)	≤ 500 meters (3)	All (4)	≤ 1,500 meters (5)	≤ 500 meters (6)	All (7)	≤ 1,500 meters (8)	≤ 500 meters (9)
Princeling firms	−0.808*** (0.025)	−0.904*** (0.034)	−0.844*** (0.033)	−0.545*** (0.035)	−0.666*** (0.043)	−0.620*** (0.043)	−0.808*** (0.030)	−0.894*** (0.040)	−0.835*** (0.038)
Princeling firms * PSCM				−0.442*** (0.037)	−0.420*** (0.048)	−0.396*** (0.049)			
Princeling firms * Retired							−0.001 (0.056)	−0.051 (0.063)	−0.044 (0.058)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Two-way clustering by firm and province	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-year-usage FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,144,507	334,232	191,896	1,144,507	334,232	191,896	1,144,507	334,232	191,896
Adjusted R-squared	0.692	0.727	0.755	0.692	0.728	0.756	0.692	0.727	0.755

Notes: Control variables include land quality, land area, sales method dummies, firm size (classified as large, medium-sized, small, or micro if its revenue in yuan is 0.3 billion or more, 30 million or more but below 0.3 billion, 3 million or more but below 30 million, or under 3 million, respectively), and ownership type; robust standard errors in parentheses. ***, **, and * indicate significance at the 0.1%, 1% and 5% levels; constant terms are not reported.

5.6 Table V: Princeling purchase and land price

Read:

- This is the paper's core table. The coefficient on princeling purchase is −0.808 in the full sample, −0.904 in the 1,500-meter matched sample, and −0.844 in the 500-meter matched sample.
- After transforming log coefficients, those imply price discounts of about 55.4%, 59.5%, and 57.0%, respectively.
- The interaction with Politburo Standing Committee connections is also strongly negative, implying an extra 17.6% to 20.7% discount on top of the baseline princeling effect.
- The retirement interaction is close to zero, which suggests that elite influence survives retirement.

5.7 Figure IV: Scatter of princeling and nearby nonprinceling prices

Read:

- Each point compares a princeling parcel's price to the average price of nearby nonprinceling parcels within 500 meters.
- Around 80% of the points lie to the left of the 45-degree line, meaning the connected transaction is cheaper than its nearby comparison.
- This is a useful visual check that the regression result is not driven by a few extreme outliers.

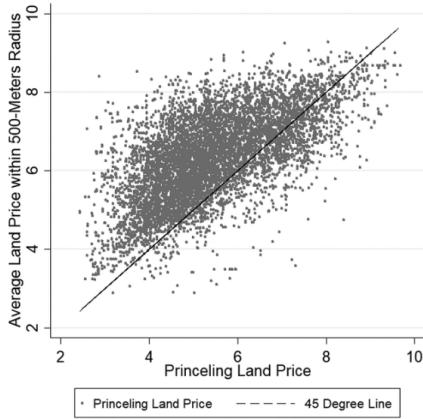


FIGURE IV
Land Prices within the 500-Meter Radius, Princeling and (Average) Nonprinceling Prices Compared

TABLE VI
QUANTITY OF LAND PURCHASED BY PRINCELING FIRMS, 2004–2016

	Log of quantity of land purchased		
	(1)	(2)	(3)
Princeling firms	0.002*** (0.000)	0.001** (0.000)	0.002*** (0.000)
Princeling firms * PSCM		0.031*** (0.004)	
Princeling firms * Retired			-0.001 (0.001)
Control variables	Yes	Yes	Yes
Clustering by firm	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Number of observations	5,690,984	5,690,984	5,690,984
Adjusted <i>R</i> -squared	0.015	0.016	0.015

Notes. Control variables include firm size (classified as large, medium-sized, small, or micro if its revenue in yuan is greater than or equal to 0.3 billion, greater than or equal to 30 million but less than 0.3 billion, greater than or equal to 3 million but less than 30 million, or less than 3 million, respectively), and ownership type; robust standard errors in parentheses; ***, **, and * indicate significance at the 0.1%, 1% and 5% levels; constant terms are not reported.

5.8 Table VI: Quantity of land purchased by princeling firms

Read:

- Connected firms also buy slightly more land: the baseline coefficient is about 0.002, which the authors interpret as roughly 0.2% more land annually.
- For firms connected to Politburo Standing Committee members, the interaction adds about 0.031, so the quantity advantage is noticeably larger.
- The quantity result is smaller than the price result, but it moves in the same direction and reinforces the access story.

5.9 Table VII: Summary statistics on political turnover

Read:

- Promotion is relatively rare for all officials, which is why any systematic increase in promotion probability is meaningful.
- The table also makes clear why the party secretary versus governor distinction matters: they are different jobs with different advancement margins.
- The descriptive statistics already hint that princeling purchase is common enough at both provincial and municipal levels to matter in the promotion analysis.

5.10 Table VIII: Princeling purchase and provincial leaders' promotion

Read:

- Provincial party secretaries who engage in princeling deals are more likely to be promoted. With controls, the ordered-probit coefficient on princeling purchase is about 0.742, corresponding to the paper's headline 23.4% increase in promotion probability.

TABLE VII
SUMMARY STATISTICS OF POLITICAL TURNOVER

	Provincial party secretaries			Provincial governors		
	Number of obs.	Mean	Std. dev.	Number of obs.	Mean	Std. dev.
Political turnover	403	100%		403	100%	
Promotion	35	8.68%		42	10.42%	
Lateral transfer or staying in office	328	81.39%		324	80.40%	
Retirement	35	8.68%		33	8.19%	
Termination	5	1.24%		4	0.99%	
Princeling purchase	399	0.596	0.491	399	0.654	0.476
Princeling discounts	403	0.685	0.686	403	0.617	0.576
Area of land purchased	401	0.904	1.286	401	0.895	1.285
Factional ties	395	0.686	0.465	403	0.591	0.492
GDP growth	403	0.137	0.064	403	0.137	0.064
Log of GDP per capita	403	7.936	0.664	403	7.936	0.664
Log of population	403	8.085	0.857	403	8.085	0.857
Tax revenue growth	390	0.172	0.102	390	0.175	0.103
Age	403	59.859	4.258	403	58.134	3.864
Age squared	403	3,601.129	495.371	403	3,394.457	439.508
Years of schooling	403	17.938	1.918	403	18.007	1.818
	Municipal party secretaries			Municipal governors		
Political turnover	3,237	100%		3,048	100%	
Promotion	209	6.46%		204	6.69%	
Lateral transfer or staying in office	2,836	87.61%		2,855	87.11%	
Retirement	177	5.47%		166	5.45%	
Termination	15	0.46%		23	0.75%	
Princeling purchase	3,663	0.497	0.500	3,663	0.480	0.500
Princeling discounts	3,597	3.232	5.322	3,605	3.224	5.318
Area of land purchased	3,657	1.704	1.468	3,661	1.579	1.287
Factional ties	3,663	0.616	0.486	3,663	0.616	0.486
GDP growth	3,511	0.151	0.076	3,511	0.151	0.076
Log of GDP per capita	3,556	9.968	0.840	3,556	9.968	0.840
Log of population	3,656	5.641	0.850	3,656	5.641	0.850
Tax revenue growth	3,090	0.150	0.105	3,090	0.150	0.105
Age	3,564	53.049	3.621	3,577	51.098	3.939
Age squared	3,564	2,827.256	381.029	3,577	2,626.469	400.419
Years of schooling	3,582	17.555	2.156	3,603	17.279	2.107

TABLE VIII
PRINCELING PURCHASE AND PROVINCIAL LEADERS' PROMOTION, 2004–2016

	Political turnover of:									
	Provincial party secretaries					Provincial governors				
	Ordered probit	Binary	Ordered probit	Ordered probit	Binary	Ordered probit	Binary	Ordered probit	Binary	Ordered probit
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Princeling purchase (= 1)	0.652*** (0.229)	0.742*** (0.262)	0.114*** (0.040)			0.018 (0.288)	−0.016 (0.314)	0.070 (0.067)		
Princeling discounts			0.814*** (0.175)						0.181 (0.175)	
Area of land purchased				0.324*** (0.084)						−0.068 (0.065)
Factional ties	0.232 (0.209)	0.011 (0.033)	0.118 (0.212)	0.302 (0.213)		−0.034 (0.208)	−0.003 (0.045)	−0.031 (0.205)	−0.052 (0.206)	
GDP growth	−1.438 (2.484)	−0.195 (0.381)	−0.444 (2.554)	−0.032 (2.530)		6.188*** (1.977)	1.250*** (0.443)	6.374*** (1.981)	6.061*** (1.983)	
Tax revenue growth	1.954 (1.230)	0.029 (0.195)	1.561 (1.241)	2.403* (1.262)		0.262 (1.070)	0.031 (0.229)	0.207 (1.058)	0.232 (1.059)	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	399	380	382	382	399	388	388	390	390	
Adjusted R-squared			0.272					0.100		

Notes: The ordinal measure of political turnover consists of four categories: termination = 0, retirement = 1, lateral transfer or staying in office = 2, and promotion = 3. The binary measure is a dummy variable that equals 1 if the official was promoted at year t , and otherwise equals 0. Control variables include tax revenue growth, log of GDP per capita, log of population size, years of education, age, and age squared. Robust standard errors are reported in parentheses. ***, **, and * indicate significance at the 0.1%, 1% and 5% levels; constant terms are not reported.

- The same is not true for governors; their coefficients are small and insignificant, while GDP growth is the variable that matters for them.
- The richer measures tell the same story: larger discounts (coefficient about 0.814) and larger land area sold (about 0.324) both predict promotion for party secretaries but not for governors.

5.11 Table IX: Princeling purchase and municipal leaders' promotion

Read:

- The provincial logic replicates one level down. Municipal party secretaries benefit from princeling-related land deals, but municipal mayors do not.

TABLE IX
PRINCELING PURCHASES AND MUNICIPAL LEADERS' PROMOTION, 2004–2016

	Political turnover of:									
	Municipal party secretaries					Municipal mayors				
	Ordered probit	Binary	Ordered probit	Ordered probit	Binary	Ordered probit	Binary	Ordered probit	Binary	Ordered probit
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Princeling purchase (= 1)	0.469*** (0.070)	0.432*** (0.077)	0.088*** (0.012)			0.079 (0.094)	0.093 (0.108)	−0.008 (0.011)		
Princeling discounts				0.100*** (0.006)					0.012 (0.008)	
Area of land purchased					0.927*** (0.066)					0.000 (0.047)
Factional ties		0.169** (0.073)	0.012 (0.011)	0.191** (0.080)	0.129 (0.082)	−0.027 (0.101)	−0.014 (0.010)	−0.021 (0.101)	−0.026 (0.101)	
GDP growth		−0.316 (0.559)	−0.099 (0.087)	−0.699 (0.616)	−0.696 (0.650)	2.798*** (0.761)	0.365*** (0.076)	2.726*** (0.761)	2.771*** (0.770)	
Tax revenue growth		−0.455 (0.398)	0.003 (0.061)	−0.588 (0.429)	−0.298 (0.444)	1.097** (0.523)	0.064 (0.052)	1.087** (0.523)	1.074** (0.524)	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	3,237	2,756	2,756	2,756	3,048	2,569	2,569	2,568	2,568	
Adjusted R-squared			0.049					0.374		

Notes: The ordinal measure of political turnover consists of four categories: termination = 0, retirement = 1, lateral transfer or staying in office = 2, and promotion = 3. The binary measure is a dummy variable that equals 1 if the official was promoted at year t , and otherwise equals 0. Control variables include tax revenue growth, log of GDP per capita, log of population size, years of education, age, and age squared. Robust standard errors are reported in parentheses. ***, **, and * indicate significance at the 0.1%, 1% and 5% levels; constant terms are not reported.

- With controls, the ordered-probit coefficient on princeling purchase for municipal party secretaries is about 0.432, and the binary promotion coefficient is about 0.088.
- Discount size and total land sold again matter for party secretaries, which fits the patron–client chain running from province to municipality.

5.12 Table X and Figure V: Xi’s campaign and the decline in price discounts

Read:

- In the matched 500-meter sample, the interaction with “transaction after 2012” is about 0.257, implying the discount shrinks by about 11.7%.
- The sharper policy variables matter even more: central inspection adds about 0.695 in the matched sample, implying a 42.6% reduction in corruption, and Xi-appointed officials add about 0.572, implying a 31.5% reduction.
- In the horse-race specification, the general post-2012 dummy becomes insignificant, while central inspections and Xi-appointed leaders remain strongly significant. That says the effective parts of the campaign were monitoring and personnel control.
- Figure V shows the time path: discounts are large and persistent before Xi, then shrink sharply after 2012 and are close to zero by 2016.

	Log of land price									
	All (1)	≤ 500 meters (2)	All (3)	≤ 500 meters (4)	All (5)	≤ 500 meters (6)	All (7)	≤ 500 meters (8)	All (9)	≤ 500 meters (10)
Princeling firms	-0.907*** (0.029)	-0.920*** (0.040)	-0.825*** (0.024)	-0.858*** (0.032)	-0.870*** (0.028)	-0.896*** (0.035)	-0.907*** (0.029)	-0.920*** (0.040)	-0.818*** (0.023)	-0.847*** (0.028)
Princeling firm *Transaction after 2012	0.318*** (0.047)	0.257*** (0.058)					0.140* (0.052)	0.093 (0.054)		
Princeling firm *Central inspection			0.819*** (0.124)	0.695*** (0.139)			0.504*** (0.079)	0.420*** (0.086)		
Princeling firm *Xi-appointed officials					0.614*** (0.055)	0.572*** (0.051)	0.449*** (0.064)	0.447*** (0.059)		
Princeling firms *Pre-2012 inspection								0.109 (0.074)	0.097 (0.070)	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Two-way clustering by firm and province	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-year-usage FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,144,507	191,896	1,144,507	191,896	1,144,507	191,896	1,144,507	191,896	1,144,507	191,896
Adjusted R-squared	0.692	0.755	0.692	0.755	0.692	0.755	0.692	0.755	0.692	0.755

Notes: Control variables include land quality, land area, sales method dummies, firm size (classified as large, medium-sized, small, or micro if its revenue in yuan is greater than or equal to 3 billion, greater than or equal to 50 million but less than 0.3 billion, greater than or equal to 3 million but less than 50 million, and less than 3 million, respectively), and ownership type and its interaction with transactions after 2012, central inspection, Xi-appointed officials, and pre-2012 inspection; robust standard errors in parentheses; ***, **, and * indicate significance at the 0.01, 0.05, and 0.10 levels; constant terms are not reported.

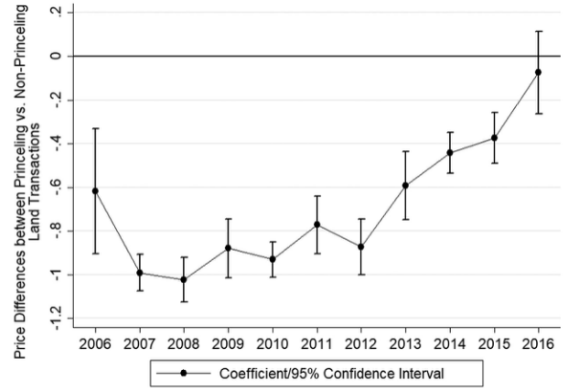


FIGURE V
Average Size of Princeling Discounts over Time

5.13 Table XI and Figure VI: Xi’s campaign and the quantity of land transferred

Read:

- Before the campaign, princeling firms buy more land; the baseline firm effect is about 0.078, or roughly an 8.1% purchase premium.
- The interactions with campaign measures are negative: about -0.022 after 2012, -0.053 under central inspection, and -0.056 when a Xi-appointed leader is in place.

- Figure VI mirrors Figure V: the quantity advantage rises before 2012 and then declines afterward, approaching zero.
- This matters because corruption here is not only about prices. It is also about allocating more land to connected buyers.

5.14 Table XII and Figure VII: Xi's campaign and the collapse of the promotion channel

Read:

- Table XII shows that Xi's campaign weakened the promotion payoff to corruption. For provincial party secretaries, the interaction of princeling purchase with post-2012 is about -1.165 , and with central inspection it is about -2.769 .
- Similar dampening appears for municipal party secretaries. The same broad pattern also holds when the paper replaces the purchase dummy with discount size or land area sold.
- Figure VII provides the sharpest visual narrative: right after the Zhou Yongkang arrest news, princeling prices rise, the price gap relative to nonprinceling deals shrinks, and the quantity purchased by connected firms falls.
- The nonprinceling market does not collapse. What changes is specifically the treatment of connected firms, which is exactly what the corruption story predicts.

TABLE XI
QUANTITY OF LAND PURCHASED BY PRINCELING FIRMS AFTER XI TOOK OFFICE

	Area of land purchased			
	(1)	(2)	(3)	(4)
Princeling firms	0.078*** (0.008)	0.073*** (0.007)	0.075*** (0.007)	0.078*** (0.008)
PP * Transactions after 2012	-0.022** (0.006)			-0.015 (0.007)
Central inspection		0.038*** (0.002)		0.025*** (0.004)
PP * Central inspection		-0.053*** (0.007)		-0.025*** (0.007)
Xi-appointed officials			0.036*** (0.002)	0.028*** (0.002)
PP * Xi-appointed officials			-0.056*** (0.006)	-0.036*** (0.006)
Control variables	Yes	Yes	Yes	Yes
Two-way clustering by firm and province	Yes	Yes	Yes	Yes
Province fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Number of observations	11,516,622	11,516,622	11,516,622	11,516,622
Adjusted R-squared	0.032	0.047	0.052	0.057

Notes. Control variables include: firm size (classified as large, medium-sized, small, or micro if its revenue in yuan is greater than or equal to 0.3 billion, greater than or equal to 30 million but less than 0.3 billion, greater than or equal to 3 million but less than 30 million, or less than 3 million, respectively), and ownership type; robust standard errors in parentheses; ***, **, and * indicate significance at the 0.1%, 1% and 5% levels; constant terms are not reported.

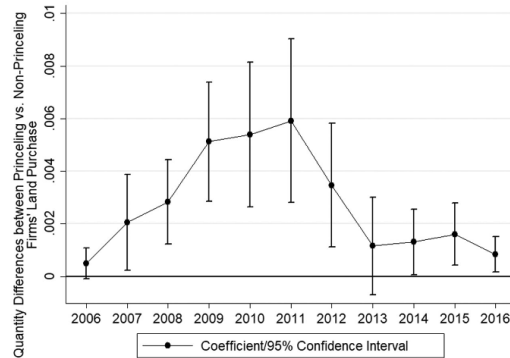


FIGURE VI
Quantity Difference of Land Purchase over Time

6 Short summary

- **Conclusion.** Chinese local officials sold land to Politburo-connected firms at extremely large discounts, and those discounts were politically rewarded.
- **Intuition.** Because local governments controlled land while higher-level elites controlled promotions, discounted land transfers became a way to buy career advancement.

- **Empirical lesson.** Transaction-level prices, close spatial matching, and institutional knowledge together can uncover corruption directly rather than inferring it indirectly from market valuation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Firms connected to princelings received enormous land-price discounts in China's primary land market.
- **Magnitude.** The matched-sample estimates imply discounts around 57% to 60%, and connections to Politburo Standing Committee members are even more valuable.
- **Reciprocity.** Provincial and municipal party secretaries who facilitated these transfers were more likely to be promoted, especially when the discounts were larger or the land area transferred was greater.
- **Campaign effect.** Xi Jinping's anticorruption campaign reduced both the price and quantity advantages enjoyed by connected firms and simultaneously reduced the promotion payoff to supplying those favors.
- **Most persuasive design feature.** The parcel-level matched-sample evidence is especially compelling because it compares very similar land parcels in very close proximity.
- **Most important limitation.** The promotion regressions remain partly observational because political connections are endogenous. They are highly suggestive, but not as airtight as the pricing evidence.

7.2 Economic intuition

- Corruption here is not a standard cash bribe from a businessman to a local official. It is closer to an intergovernmental exchange: the local official uses discretion over state assets to please families of higher-ranking politicians.
- The paper therefore links two markets that are often studied separately: the market for state assets and the market for political promotion.
- Xi's campaign worked best where it changed incentives most sharply—through inspections that increased the probability of detection and through personnel changes that altered who controlled local careers.

7.3 Takeaway

This paper shows that cronyism in China's land market was not a loose background condition but a measurable transaction-level phenomenon tied to the party hierarchy. The central insight is that decentralized control over valuable assets becomes especially dangerous when career advancement depends on pleasing a small set of national elites. The paper also shows that anticorruption can work when it changes both monitoring and personnel control, though the authors end by suggesting that deeper institutional reform would be needed to eliminate the underlying incentives.